Generative AI

Unit 1 : Assignment – Model Benchmark Challenge.

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SECTION : B
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Output Screenshots:

Experiment-1 : Text Generation

Code:

```
prompt = "The future of Artificial Intelligence is"

for name, model in models.items():

  print(f"\n{name} Output:")

  try:

    generator = pipeline("text-generation", model=model)

    print(generator(prompt, max_length=30))

  except Exception as e:

    print("Failed:", e)
```

Output Screenshot:

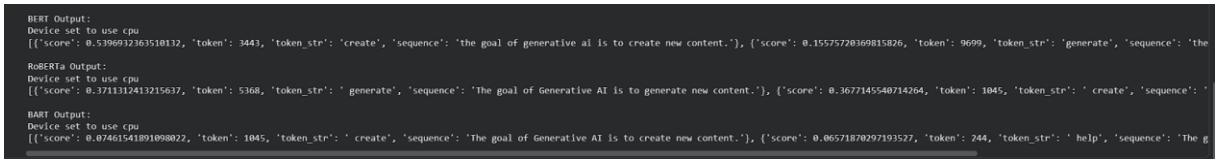
```
BERT Output:
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
If you want to use `BertLMHeadModel` as a standalone, add `is_decoder=True.`
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncation strategy. If you enc
Both `max_new_tokens` (=256) and `max_length`(=30) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_
[{'generated_text': 'The future of Artificial Intelligence is.....................................................................................................................

RoBERTa Output:
If you want to use `RobertaLMHeadModel` as a standalone, add `is_decoder=True.`
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncation strategy. If you enc
Both `max_new_tokens` (=256) and `max_length`(=30) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_
[{'generated_text': 'The future of Artificial Intelligence is'}]

BART Output:
Some weights of BartForCausalLM were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['lm_head.weight', 'model.decoder.embed_tokens.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first' truncation strategy. If you enc
Both `max_new_tokens` (=256) and `max_length`(=30) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_
[{'generated_text': 'The future of Artificial Intelligence isasuresasures vibrant vibrant vibrantocklingtonlingtonlington Wasserman clicklington vibrantlingtonlingtonarettes destructive succ typh typh typhJerific Costslingto
```

Experiment-2 : Masked Language Modeling (Fill Mask)

Code:

```
for name, model in models.items():
    print(f"\n{name} Output:")
    try:
        masker = pipeline("fill-mask", model=model)
        result = masker(sentences[name])
        print(result[:2])  # show top 2 predictions
    except Exception as e:
        print("Failed:", e)
```

Output Screenshot:



```
BERT Output:
Device set to use cpu
[{'score': 0.5396932363510132, 'token': 3443, 'token_str': 'create', 'sequence': 'the goal of generative ai is to create new content.'}, {'score': 0.1557572036981582 6, 'token': 9699, 'token_str': 'generate', 'sequence': 'the

RoBERTa Output:
Device set to use cpu
[{'score': 0.3711312413215637, 'token': 5368, 'token_str': ' generate', 'sequence': 'The goal of Generative AI is to generate new content.'}, {'score': 0.3677145540714264, 'token': 1045, 'token_str': ' create', 'sequence': '

BART Output:
Device set to use cpu
[{'score': 0.07461541891098022, 'token': 1045, 'token_str': ' create', 'sequence': 'The goal of Generative AI is to create new content.'}, {'score': 0.06571870297193527, 'token': 244, 'token_str': ' help', 'sequence': 'The g
```

Experiment-3 :Question Answering

```
question = "What are the risks?"

context = "Generative AI poses significant risks such as hallucinations, bias, and deepfakes."

for name, model in models.items():

    print(f"\n{name} Output:")

    try:

        qa = pipeline("question-answering", model=model)

        print(qa(question=question, context=context))

    except Exception as e:

        print("Failed:", e)
```

Output Screenshot:

```
BERT Output:
Device set to use cpu
Some weights of RobertaForQuestionAnswering were not initialized from the model checkpoint at roberta-base and are newly initialized: ['qa_outputs.bias', 'qa_outputs.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
{'score': 0.011541674379259348, 'start': 46, 'end': 60, 'answer': 'hallucinations'}

RoBERTa Output:
Device set to use cpu
{'score': 0.008214809466153383, 'start': 72, 'end': 82, 'answer': 'deepfakes.'}

BART Output:
Some weights of BartForQuestionAnswering were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['qa_outputs.bias', 'qa_outputs.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Device set to use cpu
{'score': 0.027998425997793674, 'start': 0, 'end': 31, 'answer': 'Generative AI poses significant'}
```

Observation table:

Task	Model	Success/Failure	Observation	Architectural Reason
Generation	BERT	Failure	Could not generate text	Encoder-only
Generation	RoBERTa	Failure	Similar to BERT	Encoder-only
Generation	BART	Success	Generated coherent text	Encoder-Decoder
Fill-Mask	BERT	Success	Correct predictions	MLM training
Fill-Mask	RoBERTa	Success	Accurate predictions	Optimized MLM
Fill-Mask	BART	Partial	Less accurate	Not MLM-based
QA	BERT	Partial	Inconsistent answers	Not QA fine-tuned
QA	RoBERTa	Partial	Similar behavior	Base model
QA	BART	Partial	Random answers	Not QA fine-tuned